# SV16 BOM - rebuild step 1 (value-level changes applied)

Source: SBSV-16.kicad_pcb (uploaded), PCB_COMPONENTS.pdf (spec), cart screenshots. Status column: BUY = not in cart, OK = covered by cart, UNDECIDED = needs a decision, PENDING = PCB footprint/value change still to do. Shared parts are listed once in the cart column.

| Spec part | Board value (step1) | Footprint (board) | Bought (cart) | Status | |
|---|---|---|---|---|---|
| LFE5U-12F-6TG144C | LFE5U-12F-6TG144C | LQFP-144 | 0 | BUY (FPGA) | |
| W25Q64JVSSIQ | W25Q64JVSSIQ | SOIC-8 5.3x6.2 | 1 | OK (have 1 of 1) | |
| W25Q64JVSSIQ | W25Q64JVSSIQ | SOIC-8 5.3x6.2 | 1 | BUY 1 more (need 2 total) | |
| Active 25 MHz XO 3.2x2.5 4-pad (ECS-3225MV-250) | 25 MHz ACTIVE XO | ASE-4Pin 3.2x2.5 (verify pads vs ECS land pattern) | 0 | BUY ECS-3225MV-250 (3 passive crystals bought - NOT usable here) | |
| 12 MHz crystal HC49/U surface mount (passive 2-pin, OK for CH340G) | 12 MHz | HC49-4H (THT) - change to SMD HC49/US | 0 | BUY 12 MHz SMD crystal; PENDING footprint change to SMD HC49 | |
| CH340G | CH340G | SOIC-16 | 1 | OK | |
| AP62300TWU-7 | AP62300TWU-7 | SOT-23-6 | 1 | OK | |
| MP1584EN | MP1584EN | SOIC-8-1EP | 1 | OK | |
| LP5907MFX-2.5 | LP5907MFX-2.5 | SOT-23-5 | 1 | OK | |
| USBLC6-2SC6 | USBLC6-2SC6 | SOT-23-6 | 4 | OK | |
| 2N7002 | 2N7002 | SOT-23 | 7 | OK | |
| MMBT3904 | MMBT3904 (changed) | SOT-23 | 9 | OK | |
| MMBT3906 | MMBT3906 (changed) | SOT-23 | 13 | OK | |
| LED green 0603 | LED green | 0603 | 7 | OK | |
| LED red 0603 (DNP by default per spec) | LED red | 0603 | 6 | OK - mark DNP on board (not done yet) | |
| LED red 0603 | LED red | 0603 | 6 | OK | |
| SS34 | SS34 | SMA | 2 | BUY 1 more (need 3 total with D9/D11) | |
| SS34 | SS34 | SMA | 2 | (see D8) | |
| SS34 (1V1 catch diode) | SS34 | SMA | 2 | (see D8) | |
| 10 uH CD54 >=1.5A | 10 uH | SWPA6045S footprint (spec: MWSA0518) - footprint NOT changed | 2 (CD54 10uH) | OK value; footprint change pending | |
| 10 uH CD54 >=1.5A | 10 uH | SWPA6045S footprint (spec: MWSA0518) - footprint NOT changed | 2 (CD54 10uH) | OK value; footprint change pending | |
| 600R @100MHz 0805 (MPZ2012S601ATD25) | 600R @100MHz | 0805 | 2 | IN CART (TDK MPZ2012S601ATD25 | qty 2); footprint change 0603 |
| 100uF 25V D-case tant | 100uF 25V tant | CP_EIA-7343-31 | 1 | OK | |

| Spec part | Board value (step1) | Footprint (board) | Bought (cart) | Status | |
|---|---|---|---|---|---|
| 22uF 63V electrolytic | 22uF 63V electrolytic | CP_Elec_6.3x5.4 | 22 | OK | |
| 470uF 25V electrolytic (CHANGED from 16V) | 470uF 25V | CP_Elec_8x10.5 | 0 | BUY 470uF 25V (16V stock not used for C23; derating 75% too high) | |
| 220uF 25V hybrid | 220uF 25V hybrid | CP_Elec_8x10.5 | 11 | OK | |
| 10uF 0805 ceramic (spec) -> 10uF 35V electrolytic (decision) | 10uF | C_0805 - footprint NOT changed | 11 (10uF 35V electrolytic) | footprint change pending (CP_Elec_4x5.4) | |
| 10uF 0805 ceramic (spec) -> 10uF 35V electrolytic (decision) | 10uF | C_0805 - footprint NOT changed | 11 | footprint change pending | |
| 10uF electrolytic (decision) | 10uF 35V electrolytic | CP_Elec_4x5.4 | 11 | OK | |
| 22uF electrolytic (decision) | 22uF 63V electrolytic | CP_Elec_6.3x5.4 | 22 | OK | |
| 22uF electrolytic (decision) | 22uF 63V electrolytic | CP_Elec_6.3x5.4 | 22 | OK | |
| 10uF electrolytic (decision) | 10uF 35V electrolytic | CP_Elec_4x5.4 | 11 | OK | |
| 470nF 0603 (spec) | 470nF | C_0805 - footprint NOT changed | 20 (470nF 0603) | footprint change pending (0603) | |
| not in spec | 100nF | 0603 | 40 | KEEP board value (not in spec, purpose unverified) - report | |
| not in spec | 100pF | 0603 | 0 | KEEP board value (not in spec, purpose unverified) - report | |
| 22pF C0G 0805 50V 5% (YAGEO CC0805JRNPO9BN220) | 22pF | 0603 - change to 0805 | 0 | BUY 22 pF 0805 C0G; PENDING footprint change 0603->0805 | |
| 22pF C0G 0805 50V 5% (YAGEO CC0805JRNPO9BN220) | 22pF | 0603 - change to 0805 | 0 | BUY (see C36); PENDING footprint change 0603->0805 | |
| 100nF 0603 | 100nF | 0603 | 40 (shared) | OK | |
| 100nF 0603 | 100nF | 0603 | (shared) | OK | |
| 470nF 0603 | 470nF | 0603 | (shared) | OK | |
| 100nF 0603 | 100nF | 0603 | (shared) | OK | |
| 100nF 0603 | 100nF | 0603 | (shared) | OK | |
| 10k 0603 5% (YAGEO RC0603JR-0710KL) | 10k | 0603 | 0 | BUY 10k 0603 5% (Yageo RC0603JR-0710KL); in cart (qty 110) | |
| 4.7k 0805 | 4.7k (changed) | 0805 | (4.7k 0805 pack) | OK | |
| 4.7k 0805 (from bought pack) | 4.7k | 0603 - change to 0805 | (4.7k 0805 pack) | PENDING footprint change 0603->0805; use bought 4.7k 0805 pack | |
| 1k 1206 (from bought pack) | 1k | 0603 - change to 1206 | (1k 1206 pack) | PENDING footprint change 0603->1206; use bought 1k 1206 pack | |
| 100R 1206 (spec) - board 1206 | 100R (changed) | 1206 | (100R 0805 pack bought) | footprint change to 0805 pending (bought 0805) | |

| Spec part | Board value (step1) | Footprint (board) | Bought (cart) | Status | |
|---|---|---|---|---|---|
| 10k 0603 5% (YAGEO RC0603JR-0710KL) | 10k | 0603 | 0 | BUY 10k 0603 5% (Yageo RC0603JR-0710KL), 7 pieces | |
| 0R | 0R (changed) | 0603 | 0 | solder-bridge option (no purchase needed) | |
| 4.7k 0805 (from bought pack) | 4.7k | 0603 - change to 0805 | (4.7k 0805 pack) | PENDING footprint change 0603->0805; use bought 4.7k 0805 pack | |
| 10k 0603 5% (YAGEO RC0603JR-0710KL) | 10k | 0603 | 0 | BUY 10k 0603 5% (Yageo RC0603JR-0710KL) | |
| 470R 0805 | 470R (changed) | 0805 | (470R 0805 pack) | OK | |
| 1k 1206 | 1k (changed) | 1206 | 1 (1k 1206 pack) | OK | |
| 10k 0603 5% (YAGEO RC0603JR-0710KL) | 10k | 0603 | 0 | BUY 10k 0603 5% (Yageo RC0603JR-0710KL) | |
| 1k 1206 | 1k (changed) | 1206 | (1k 1206 pack) | OK | |
| 1k 1206 | 1k (changed) | 1206 | (1k 1206 pack) | OK | |
| 5.1k 0805 | 5.1k (changed) | 0805 | (5.1k 0805 pack) | OK | |
| 5.1k 0805 | 5.1k (changed) | 0805 | (5.1k 0805 pack) | OK | |
| 12.4k 0402 | 12.4k (changed) | 0402 | 30 | OK | |
| 33k 1206 | 33k (changed) | 1206 | (33k 1206 pack) | OK | |
| 100k 0603 | 100k (changed) | 0603 | 11 (thin film) | OK | |
| 100k 0603 | 100k (changed) | 0603 | (shared) | OK | |
| 100k 0603 | 100k (changed) | 0603 | (shared) | OK | |
| not in spec | R44 (no value) | 0603 | - | LEAVE UNSET (no value anywhere) - report | |
| not in spec | 100k | 0603 | (shared) | KEEP board value (not in spec, purpose unverified) - report | |
| 33k 1206 | 33k (changed) | 1206 | (shared) | OK | |
| 10k 0603 1% (YAGEO RC0603JR-0710KL, 5%) | 10k | 1206 - change to 0603 | 0 | BUY 10k 0603 5% (Yageo RC0603JR-0710KL); footprint change 1206->0603 | |
| AP62300 EN divider (value not given in spec) | 100k | 0603 | (shared) | KEEP board value (not in spec, purpose unverified) - report | |
| AP62300 EN divider (value not given in spec) | 33k | 1206 | (shared) | KEEP board value (not in spec, purpose unverified) - report | |
| Standard 2.54 mm straight male pin header, 1x10 (most available). Footprint unchanged. | JTAG | 2.54mm 1x10 | 0 | USER HAS (not in screenshots) - confirm | |
| Spec: 1.27 mm 2x5 box header. Less common; leave unpopulated and use J1 (spec option). | JTAG-1.27 | 1.27mm 2x5 | 0 | OPTIONAL | |

| Spec part | Board value (step1) | Footprint (board) | Bought (cart) | Status | |
|---|---|---|---|---|---|
| Standard 2.54 mm straight male pin header, 2x10 (most available). Footprint unchanged. | EXP-A | 2.54mm 2x10 | 0 | USER HAS (not in screenshots) - confirm | |
| Standard 2.54 mm straight male pin header, 2x10 (most available). Footprint unchanged. | EXP-B | 2.54mm 2x10 | 0 | USER HAS (not in screenshots) - confirm | |
| Standard 2.54 mm straight male pin header, 1x10 (most available). Footprint unchanged. | EXP-C | 2.54mm 1x10 | 0 | USER HAS (not in screenshots) - confirm | |
| Shrouded keyed 1x6 2.54 mm header (motor connector) | MOTOR | 2.54mm 1x6 | 0 | BUY shrouded keyed 1x6 2.54 mm (or confirm you have one) | |
| Standard 2.54 mm straight male pin header, 2x25 (footprint only, not fitted). | absent | - | 0 | ADD footprint (not fitted) | |
| USB-C 16-pin HRO | USB-C | HRO TYPE-C-31-M-12 | 4 | OK | |
| Barrel jack 5.5/2.1 | 7-12 V | CUI PJ-102AH | 0 | USER HAS (not in screenshots) - confirm | |
| Standard 2.54 mm straight male pin header, 1x4 (most available). Footprint unchanged. | CONSOLE | 2.54mm 1x4 | 0 | USER HAS (not in screenshots) - confirm | |
| Standard 2.54 mm straight male pin header, 1x2, with shunt (most available). Footprint unchanged. | DTR ENABLE | 1x02 | 0 | USER HAS (not in screenshots) - confirm | |
| 6x6 tact | RESET | SW_PUSH_6mm | 0 | USER HAS (not in screenshots) - confirm | |
| 6x6 tact | RECONFIG | SW_PUSH_6mm | 0 | USER HAS (not in screenshots) - confirm | |
| 1 mm pads | OK | TestPoint | - | OK | |
| M3 mount holes | OK | - | - | OK | |